# Shortcuts and Temporal Leakage in the Bangalore EEG Epilepsy Dataset: a Provenance Audit and Leakage-Aware Benchmark of Lightweight Seizure-Type Classifiers

*[Author names and affiliations removed for double-blind review]*

*Track 5: Digital Life Sciences, Health, Agriculture & Biodiversity (healthcare artificial intelligence)*

## Abstract

Public electroencephalography (EEG) datasets are widely used to develop seizure classifiers, and the recently released Bangalore EEG Epilepsy Dataset (BEED) has quickly attracted reports of 95–99% accuracy. We asked whether these figures reflect physiological discrimination or properties of how the public file was assembled. We audited the 8000-row, 16-column file with autocorrelation, change-point and exact-matching analyses, and benchmarked six classical and four small deep-learning models under a random row-level split and under a leakage-aware protocol (contiguous time blocks with a purge gap, nested tuning, a pre-specified primary model, rolling-origin evaluation, cluster-bootstrap intervals and Holm-adjusted corrected t-tests). All 2000 'healthy' rows are identical to 16-sample excerpts of the single-channel Bonn eyes-closed recordings, whereas the seizure classes behave as 16-channel time points of continuous recordings, so consecutive rows are near-duplicates. The best random-split macro-F1 was 0.98. Under the leakage-aware protocol, the pre-specified model (LightGBM on raw values) reached 0.75 ± 0.08 for four classes and 0.68 ± 0.10 for the three seizure types in rolling-origin evaluation (single hold-out 0.81 and 0.75). The healthy class remained almost perfectly separable (per-class F1 ≥ 0.99), consistent with a source shortcut. Because each seizure class appears to be a single recording, a per-channel amplitude fingerprint alone reproduced about 90% of the above-chance performance (0.72

and 0.65): the leakage-aware score measures within-recording temporal extrapolation, not seizure-type generalisation. Results on this file require leakage-aware, pre-specified evaluation, and neither the healthy class nor the seizure-type labels support clinical claims. We release a reproducible pipeline.

**Keywords:** electroencephalography; epileptic seizure classification; data leakage; dataset provenance; blocked cross-validation; rolling-origin evaluation; shortcut learning; explainable machine learning

## 1 Introduction

Epilepsy diagnosis relies heavily on visual review of the electroencephalogram (EEG), which is slow and requires specialist expertise; machine-learning (ML) classifiers are therefore an active research area [1–4]. Progress depends on public datasets such as the Bonn recordings [5] and the CHB-MIT scalp database [1,6]. The Bangalore EEG Epilepsy Dataset (BEED), released through the UCI Machine Learning Repository in 2024 [7,8], adds a four-class problem: healthy subjects (class 0), generalised seizures (1), focal seizures (2) and seizure events accompanied by behaviours such as blinking or staring (3). Distinguishing focal from generalised onset matters clinically because it guides treatment choices, and seizure-type classification is markedly harder than seizure detection [9].

Within two years, several studies reported 95–99% accuracy on BEED [8,10–17]. However, the public BEED file contains only 8000 rows of 16 integers with no subject or recording identifiers, which cannot be reconciled with its documentation (80 subjects, 16,000 20-s segments). Most reported evaluations split rows at random. In EEG and other autocorrelated data, random splits can place almost identical neighbouring samples in training and test sets, inflating performance [18–20]. How much this applies to BEED had not been quantified.

**Research questions.** We addressed four questions. (RQ1) What do the rows and columns of the public BEED file represent, and is the representation the same across classes? (RQ2) How

much of the reported performance survives an evaluation that prevents temporal leakage? (RQ3) Once shortcuts are controlled, which signal properties separate generalised, focal and seizure-event recordings, and are the predicted probabilities calibrated? (RQ4) What is the computational footprint of the candidate models on a general-purpose CPU?

**Contributions.** (i) A structural and provenance audit showing that the healthy class is identical to excerpts of the Bonn eyes-closed set whereas the seizure classes behave as multichannel time points. (ii) A leakage-aware protocol (grouped, temporally blocked splits with a purge gap, nested tuning, rolling-origin evaluation and a pre-specified primary model) and a quantification of the gap between random and blocked evaluation. (iii) Fingerprint and normalisation baselines showing that most of the leakage-aware signal is recording-specific. (iv) A benchmark of ten models with two interval types, Holm-adjusted corrected resampled t-tests, calibration, Shapley-value interpretation and a footprint profile. (v) A reproducible pipeline (Supplementary File S1).

## 2 Related Work

### 2.1 Seizure detection and classification benchmarks

The Bonn dataset [5] contains five sets of 100 single-channel, 23.6 s segments sampled at 173.61 Hz; a widely redistributed UCI version cuts each segment into 1 s chunks of 178 samples and shuffles them [21]. Deep convolutional networks report near-perfect accuracy on Bonn [2]; CHB-MIT provides long multichannel paediatric recordings at 256 Hz [1,6] used for detection and prediction [3,4]. Classic hand-crafted EEG descriptors include Hjorth parameters [22], line length [23], sample and permutation entropy [24,25] and discrete wavelet energies [26]. For seizure-type classification on the TUH corpus, the best weighted F1 was 0.901 under five-fold seizure-wise cross-validation (k-nearest neighbours, TUSZ v1.4.0) but only 0.561 under three-fold patient-wise cross-validation (XGBoost, TUSZ

v1.5.2) [9]; the settings also differ in corpus version and classifier, but the gap shows how strongly the protocol shapes conclusions.

### 2.2 Prior work on BEED

BEED was released through the UCI repository in 2024 [7], and its creators have reported 96–99% accuracy with spectral and multi-domain features and boosting or stacked classifiers [8,15,17]. Table 1 summarises the published studies. Two points matter when comparing them. First, several headline figures are not four-class results. The 96.71% of SeqBoostNet [8] is the mean of six pairwise binary tasks, ranging from 91.16% (focal vs seizure events) to 99.83% (focal vs healthy). The 99% of [15] refers to seizure-versus-healthy detection. Second, the studies that evaluated on the public file [8,10–12] used random, stratified or otherwise row-level splits, sometimes after fitting UMAP on all rows before splitting [8]. A structured search found nine further 2025–2026 reports, mostly conference papers (Supplementary Table S19); where their abstracts give a figure it is similarly high (accuracy, F1 or AUC of 95–100%), and none describes a grouped or temporally blocked split.

**Table 1. Published BEED studies: task, evaluation protocol as implemented, and reported result**

| Study | Task | Evaluation protocol (as implemented) | Reported result |
|---|---|---|---|
| Najmusseher et al. [15] (dataset creators) | Seizure vs healthy; seizure onset vs events; seizure types | Not stated in the accessible abstract; described as sample-level by [13] | Accuracy 99% (seizure vs healthy) |
| Najmusseher & Nizar Banu [8] (SeqBoostNet) | Six pairwise binary tasks; no 4-class task | UMAP and FFT features computed on all rows, then a row-level split | Accuracy 91.16–99.83% (mean of tasks 96.71%) |
| Paneru [10] | 4-class | UMAP + FFT; row-level hold-out; 10-fold nested CV | Accuracy 97.06% (hold-out); 91.51% (nested CV) |
| Nandatama [11] | 4-class | Stratified random 80/20 row split | Weighted F1 0.9831 |
| Sajid et al. [12] (PaperNet) | 4-class | Stratified random 70/15/15 sample-level split (abstract: “subject-independent”) | Macro-F1 0.9576 |
| Obaido & Esenogho [13] | Binary | GroupKFold by subject on data described as 10 subjects, 19 channels, 1-s windows; matches no public BEED file | TabPFN accuracy 0.997 |
| Heba & Atya [14] (EpilepsyFormer) | Binary | Subject-independent 5-fold CV (paper) on 14-channel 20-s segments that the public CSV cannot yield; released code: random 80/20 split | Accuracy 98.3% |
| Shah & Toma [16] | Binary (class 3 dropped) | 70/15/15 split of 60 pseudo-patients built from 100-row blocks in file order | Accuracy 97.56% |

UMAP, uniform manifold approximation and projection; FFT, fast Fourier transform; CV, cross-validation. Protocols are as implemented in each paper's methods or code, which sometimes differs from the abstract. The public UCI file (BEED_Data.csv; doi:10.24432/C5K33B) has 8000 rows × 16 integer columns, 2000 rows per

class and no subject or recording identifiers. Data described, sources consulted and details: Supplementary Table S18; further 2024–2026 reports: Supplementary Table S19.

Three studies report subject- or patient-independent results [13,14,16], but none is a subject-independent evaluation of the public file (Table 1). The first two describe data (subject identifiers, 19 or 14 channels, 1-s or 20-s waveforms) that do not match the only file on the UCI page, which their data statements nevertheless cite, and the released code of the second splits segments at random; the third built “patients” from 100-row blocks in file order. We found no public BEED release with these properties: the UCI file and two Kaggle redistributions are all named BEED_Data.csv and have the same size (404,412 bytes). All three studies are also binary tasks, a contrast that on the public file includes the Bonn-derived healthy class, which a single threshold on within-row lag-1 autocorrelation separates with blocked-protocol accuracy 0.997 (Section 4.4).

### 2.3 Leakage in machine-learning evaluation

Data leakage, in which information about the test set reaches the model during training, is a leading cause of irreproducible ML results [18]. In EEG, splitting segments rather than subjects inflated deep-learning accuracy substantially [19], and for autocorrelated data blocked cross-validation with buffers is recommended [20]. PaperNet [12] noted that each row of the public file is a single 16-electrode time point but still used a stratified random split, and Shah and Toma [16] grouped rows into 100-row “patients” without a purge gap. To our knowledge, no BEED study has (i) traced the origin of any class, (ii) shown that the row layout differs between classes, or (iii) quantified how row adjacency inflates random-split performance and benchmarked models under a temporally blocked, purged protocol.

## 3 Materials and Methods

The study design (Supplementary Fig. S10) comprises a structural and provenance audit, provenance-based grouping, and two evaluation protocols sharing identical features, models,

tuning and metrics, supplemented by a rolling-origin evaluation and recording-fingerprint baselines.

### 3.1 Dataset

We used the BEED CSV file from the UCI repository [7] (8000 rows; integer columns X1–X16; label y). Its documentation (retrieved 8 October 2026) describes 16,000 20-s segments recorded with the 10–20 system at 256 Hz from 80 adults (20 per category), 16 channels with a binary seizure label, 200 time points per subject, and 8000 instances in four classes. These statements are mutually inconsistent and do not match the file, which has 8000 rows, four label values and no subject column; 2000 rows per class at 200 time points per subject would be 10 subjects per class, and a 20-s segment at 256 Hz would require 5120 samples (verbatim quotations in Supplementary Table S20). Each class has 2000 contiguous rows (label changes at rows 2000, 4000 and 6000), with no missing values or duplicated rows. No filtering, resampling or artefact removal was applied; standardisation was fitted on training data only.

### 3.2 Structure, provenance and change-point analysis

For each class we computed within-row (column k vs k+l) and across-row (row i vs i+l) correlations and tested whether the within-row correlation depends only on lag (Toeplitz, as for time samples) or on position (as for electrodes). Because class-0 rows were smooth and lacked across-row continuity, we searched every 16-sample window of every row of the UCI Bonn file [21] for exact integer matches with BEED rows (the SHA-256 of the file is recorded in the code). To test the documented layout of 200 time points per subject, we looked for splice points in each seizure class using large steps (norm above eight times the class median), penalised kernel change-point detection (PELT) on the z-scored signal and its first differences, and comparisons of step sizes and spatial correlations at 100- and 200-row boundaries against a random-placement null (Supplementary Section S7.2). Finally, we

searched CHB-MIT v1.0.0 [1,6] for the source of classes 1–3 with a gain-, offset- and channel-order-invariant normalised cross-correlation, validated by a positive control (Supplementary Section S7.5).

### 3.3 Evaluation protocols

**Random protocol.** A stratified 80/20 row-level hold-out and stratified five-fold cross-validation (CV) repeated twice (5 × 2), as in most prior work. **Blocked protocol.** Class-0 rows were grouped by their Bonn source segment (100 groups) and class-1–3 rows into contiguous 200-row blocks (0.78 s at 256 Hz). The hold-out test set comprised the final 20% of each seizure recording and 20 randomly chosen Bonn segments; CV folds assigned whole groups to folds within each class. Training rows within 50 rows of any test row of the same class were purged (across-row autocorrelation is close to zero at this lag; Table 2). Bonn set B comes from five volunteers, so grouping class 0 by segment slightly favours that class. **Rolling origin.** Because the final 20% is a single test-block position, fold k (k = 0–4) tested rows 400k to 400k + 399 of every seizure class plus a fixed fifth of the Bonn segments, with a 50-row purge on both sides, so that every part of every recording was tested once. In all protocols, classical hyper-parameters were chosen by an inner three-fold CV following the outer protocol (nested CV), optimising macro-F1. **Pre-specified primary model.** The primary model was the configuration (model × feature set) with the highest mean macro-F1 in the nested purged blocked 5 × 2 CV on the development data; no test block was used. As a check against selection effects, we also ran fully nested selection inside the rolling origin. We studied three tasks: four classes; healthy versus seizure; and the three seizure types (generalised, focal, seizure events), which excludes the healthy class and so removes the source shortcut.

### 3.4 Feature engineering

Each row was described by 57 engineered features: time domain (24; moments, quantiles, root mean square, Hjorth parameters [22], line length [23], difference statistics, zero crossings, extrema and lag-1 autocorrelation), frequency domain (14; relative power in the eight non-DC bins of a Hann-windowed 16-point FFT, spectral entropy, centroid, 80% edge, peak bin and a low/high power ratio, because conventional band powers cannot be resolved from 16 points), nonlinear (7; permutation [25], sample and approximate [24] and singular-value-decomposition entropy; Petrosian, Katz and Higuchi fractal dimensions) and wavelet (12; statistics of the Daubechies-2 level-2 sub-bands [26,27]). Feature sets were raw (16), engineered (57) and raw+engineered (73). For class 0 these features are temporal; for classes 1–3 they describe the spatial profile across electrodes (Section 4.1).

### 3.5 Models and training

Classical models [28]: multinomial logistic regression (LR), k-nearest neighbours (kNN), radial-basis-function support vector machine (SVM-RBF), random forest (RF), XGBoost [29] and LightGBM [30], plus a majority-class baseline; compact grids are listed in Supplementary Table S1. Deep models [31] took the 16 standardised raw values as input: a multilayer perceptron (MLP), a three-layer one-dimensional convolutional network (1D-CNN), a bidirectional long short-term memory network (BiLSTM) and a two-layer transformer encoder; the last three treat the 16 electrodes, in column order, as a sequence. They were trained with AdamW (weight decay $10^{-4}$, batch 128, $\leq$ 100 epochs, early stopping with patience 15) and tuned over a 12-point grid (learning rate, width, dropout) by a single purged blocked inner hold-out (80/20 of the outer-training groups) in every blocked outer fold, then refitted on the whole outer-training set. Under the random protocol they used fixed default hyper-parameters (learning rate 0.002, $\leq$ 60 epochs, patience 8). All random seeds were fixed (seed 42).

### 3.6 Recording-fingerprint and normalisation baselines

Because each seizure class may be a single recording, a classifier can succeed by recognising recording-specific amplitude and offset rather than seizure type. We therefore trained tuned LR and LightGBM models on fingerprint features only (per-channel absolute value |x|; row standard deviation; row mean; four row statistics), and retrained LR, kNN, SVM-RBF and LightGBM on raw values after (a) per-recording channel-wise z-scoring (recording = class for classes 1–3, Bonn segment for class 0; a transductive diagnostic) and (b) per-row normalisation (label-free). All normalised inputs were first de-quantised with uniform noise on (−0.5, 0.5), because z-scoring integer data class by class places each class on its own value lattice, which tree models can exploit. All baselines used the same protocols as the main models.

### 3.7 Metrics, uncertainty and statistical testing

We report accuracy, macro-F1, Cohen's kappa, one-vs-rest macro area under the ROC curve (AUC), per-class precision and recall, Brier score and expected calibration error (ECE, 15 bins) [32]. Uncertainty was quantified in two ways: a class-stratified cluster bootstrap (2000 resamples) of the final-20% hold-out over Bonn segments (class 0) or 50-row chunks (classes 1–3), which was insensitive to chunk size (Supplementary Table S32; naive row-level intervals: Table S4); and the spread over the five rolling-origin test blocks (mean ± $t_4 \cdot SD/\sqrt{5}$ and min–max), which captures variance due to test-block placement that the bootstrap omits. Models were compared on the 10 outer CV folds with the Nadeau–Bengio corrected resampled t-test [33,34] against the best model in each task × protocol setting, Holm-corrected across all four settings; a pre-specified family of four tests compared the primary model with the LR(|x|) fingerprint baseline in blocked CV and rolling origin. Friedman tests with Nemenyi post-hoc analysis [35] assume independent folds and are descriptive only.

### 3.8 Interpretability, ablation, temporal context and computational footprint

For LightGBM on raw+engineered features with fixed default hyper-parameters we computed TreeSHAP values [36] on the hold-out set (a descriptive model, not the primary model), and a feature-group ablation retrained it on each group alone and on all engineered features minus each group. To check separability given true temporal context, multichannel windows of classes 1–3 (32 or 64 rows; per-channel Welch band powers and line length) were classified by an LR tuned by inner purged blocked CV, on the same splits as single rows and with a paired single-row SVM-RBF comparator. For the computational footprint we measured the median of 500 single-sample calls after 30 warm-up calls on one CPU thread and the size of the serialised models (native and ONNX).

### 3.9 Software and hardware environment

Experiments ran on a Linux virtual machine (8 virtual Intel Xeon CPUs, 16 GB RAM, no GPU) with Python 3.13, scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0, PyTorch 2.14.1 (CPU), SHAP 0.52.0, SciPy 1.18.1, PyWavelets 1.10.0, AntroPy 0.2.2, ruptures (change-point detection) and ONNX Runtime. No wearable or embedded hardware was used. A single command (run_all.py) regenerates the core analyses, and the scripts in extended_analyses/ regenerate the remaining analyses (Supplementary File S1).

## 4 Results

### 4.1 The columns mean different things in different classes

Class 0 showed the signature of time samples: the within-row lag-1 correlation was 0.930 at every position (standard deviation across positions 0.002), decaying smoothly with lag, whereas consecutive rows were uncorrelated (r = -0.001) (Fig. 1, Table 2). Classes 1–3 showed the opposite pattern: across-row lag-1 correlations of 0.955, 0.845 and 0.983, falling to about zero by 50 rows, with position-dependent, non-Toeplitz correlations between

columns. The strongest correlations occurred at lags 4, 8 and 12, a pattern consistent with four chains of four electrodes, as in a longitudinal bipolar montage. Thus, in classes 1–3, each row behaves as one time point of a 16-channel recording. Classes also differed sharply in amplitude (Table 2).

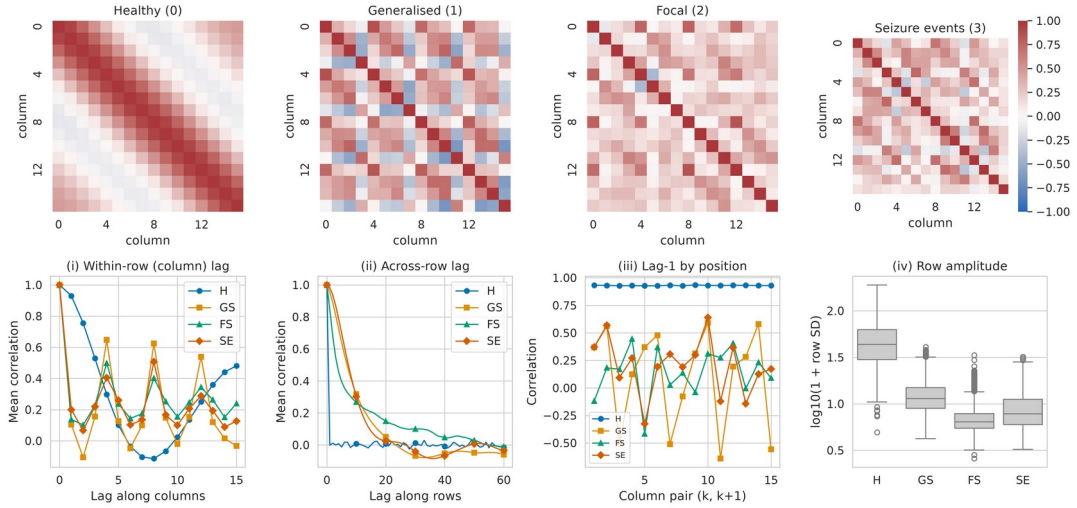



**Figure 1. Correlation structure reveals different data layouts across classes.** Top: column-by-column correlation matrices per class. Class 0 shows a Toeplitz band (time samples); classes 1–3 show position-specific structure with strong correlations at lags 4, 8 and 12 (electrode chains). Bottom: (i) mean within-row correlation versus column lag; (ii) mean across-row correlation versus row lag; (iii) lag-1 correlation for each adjacent column pair; (iv) distribution of row standard deviation (log scale). Classes: H, healthy; GS, generalised seizure; FS, focal seizure; SE, seizure events.

**Table 2. Structural audit of the public BEED file by class**

| Class | Row SD | Within-row lag-1 r | SD of lag-1 r over positions | Across-row lag-1 r | Across-row lag-50 r | Exact Bonn matches | Inferred row layout |
|---|---|---|---|---|---|---|---|
| H (0) | 70.0 | 0.930 | 0.002 | -0.001 | 0.012 | 2000 | 16 samples × 1 channel |
| GS (1) | 15.1 | 0.105 | 0.433 | 0.955 | -0.047 | 0 | 1 time point × 16 channels |
| FS (2) | 7.9 | 0.138 | 0.219 | 0.845 | 0.030 | 0 | 1 time point × 16 channels |
| SE (3) | 10.2 | 0.200 | 0.249 | 0.983 | 0.007 | 0 | 1 time point × 16 channels |

SD, standard deviation (dataset units); r, Pearson correlation. Exact matches: rows identical to a 16-sample window of the UCI Bonn file (2000 rows per class).

The change-point analysis did not support the documented layout of 200 time points per subject (Supplementary Section S7.2, Fig. S12). Kernel change-point detection found no

change in the increment process at penalties of 5–20 × d·ln n; the few signal-level mean shifts (class 1, row 307; class 2, rows 418, 488, 599) lay off 100- and 200-row boundaries; steps at these boundaries were not larger than ordinary steps (random-placement $p \geq 0.33$); and discontinuities were not enriched at boundaries (hypergeometric $p \geq 0.12$). Each seizure class is therefore consistent with one, or a few seamlessly joined, non-stationary recordings of about 7.8 s at 256 Hz. One candidate discontinuity (class 1, row 1900) cannot be excluded; the rolling origin tests it once.

### 4.2 The healthy class is identical to excerpts of the Bonn dataset

All 2000 class-0 rows matched a 16-sample window of the UCI Bonn file exactly and uniquely; no row from classes 1–3 matched (Fig. 2). Every match came from Bonn label 4 (set B, healthy volunteers, eyes closed [5,21]), always at the same position (samples 159–174 of the 178-sample chunk), from 2000 distinct chunks of 100 distinct segments, in the same order as in the UCI file. Class 0 therefore appears to be taken from the Bonn set-B data (single-channel EEG sampled at 173.61 Hz at another centre), not from 16-channel EEG recorded in Bangalore; a class-0 row spans about 92 ms of one channel, whereas a seizure-class row is a single instant (3.9 ms at 256 Hz), and the oscillation in its within-row correlation (Fig. 1) matches the eyes-closed alpha rhythm at that rate. A search of CHB-MIT for the source of classes 1–3 was negative (Section 7).

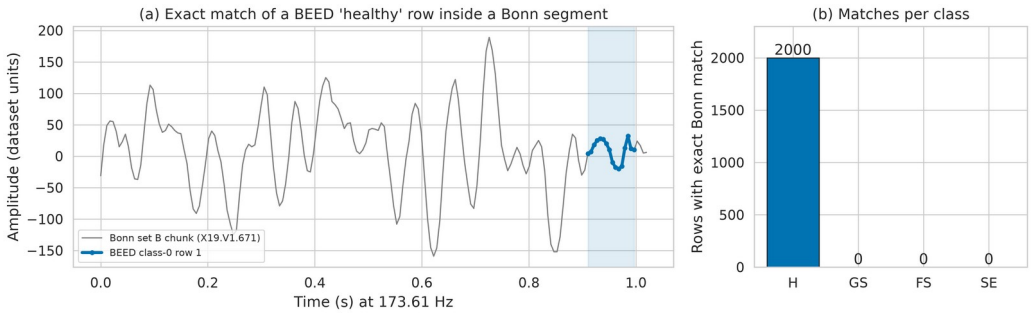



**Figure 2. BEED healthy rows are identical to excerpts of Bonn eyes-closed recordings.** (a) A Bonn set-B one-second chunk (grey, 173.61 Hz) with the identical BEED class-0 row

overlaid (blue). (b) Number of rows per BEED class with an exact match anywhere in the 11 500 × 178 UCI Bonn file.

### 4.3 Quantifying temporal leakage

Under the random split, the nearest training neighbour of a test row in classes 1–3 was the immediately adjacent row of the same recording in 87.8%–95.2% of cases (median distance one row), versus 0.0% for class 0. Fig. 3 shows how macro-F1 declines as rows are allocated to folds in increasingly long contiguous blocks: LightGBM fell from 0.976 (random) to 0.759 (200-row blocks with purge) and 1-nearest-neighbour from 0.990 to 0.703. The decline tracks the decay of the across-row autocorrelation, so near-perfect random-split scores depend largely on near-duplicate neighbours.

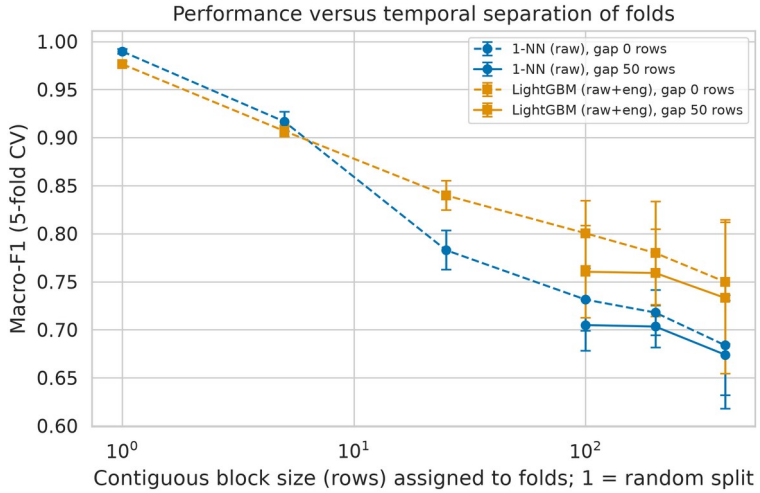



**Figure 3. Performance falls as folds are separated in time.** Five-fold cross-validation macro-F1 (mean ± SD) when rows of classes 1–3 are assigned to folds in contiguous blocks of the given size (block size 1 is a random split). Dashed lines: no purge; solid lines: 50-row purge (shown for blocks of at least 100 rows, because smaller blocks leave almost no purged training data). LightGBM: raw+engineered features, fixed hyper-parameters.

### 4.4 Leakage-aware performance of the pre-specified model

The selection rule picked LightGBM on raw values in both tasks (blocked CV 0.777 ± 0.048 for four classes, runner-up MLP (raw) 0.774; 0.720 ± 0.053 for three classes, runner-up XGBoost (raw) 0.716). Table 3 and Supplementary Fig. S13 give its leakage-aware estimates. In the rolling origin, which we take as the primary estimate, four-class macro-F1 was 0.748 ± 0.080 (t-interval 0.648–0.847; range over test blocks 0.631–0.839) and three-class macro-F1 0.677 ± 0.104 (0.548–0.805; range 0.534–0.798). Fully nested selection inside the rolling origin gave almost the same values (0.747 ± 0.080 and 0.665 ± 0.105). On the single final-20% hold-out, the primary model reached 0.810 (cluster-bootstrap CI 0.771–0.845; accuracy 0.816, kappa 0.755, AUC 0.949) and 0.752 (0.694–0.806).

The final 20% is one of the easier test blocks for every model (Supplementary Fig. S13), so every hold-out interval sits in the upper part of the rolling-origin range, and the cluster bootstrap (width about 0.07–0.11) is narrower than the variation due to test-block position (rolling-origin t-intervals 0.20 and 0.26 wide). Selecting the best of about 16 classical configurations on this favourable block would have given 0.842 (CI 0.800–0.880) and 0.776 (0.712–0.832) (SVM-RBF on raw values), whereas in blocked CV and rolling origin the same SVM-RBF reached 0.755 ± 0.046 and 0.754 ± 0.085 (four classes) and 0.685 ± 0.073 and 0.689 ± 0.090 (three classes). A single hold-out also mis-ranks models: the tuned transformer scored 0.861 on the four-class hold-out but 0.749 in blocked CV and 0.747 in the rolling origin.

**Table 3. Macro-F1 of the pre-specified primary model under random and leakage-aware evaluation**

| Evaluation | Four classes | Three seizure types |
|---|---|---|
| Random-split 5 × 2 CV, best model (kNN, raw) | 0.984 ± 0.004 | 0.983 ± 0.006 |
| Random-split hold-out, primary model | 0.986 | 0.968 |
| **Primary model (LightGBM, raw), leakage-aware** | | |
| Rolling origin, mean ± SD (primary estimate) | **0.748 ± 0.080** | **0.677 ± 0.104** |
| Rolling origin, t-interval [min–max] | 0.648–0.847 [0.631–0.839] | 0.548–0.805 [0.534–0.798] |
| Fully nested selection inside rolling origin | 0.747 ± 0.080 | 0.665 ± 0.105 |
| Purged blocked 5 × 2 CV (selection data) | 0.777 ± 0.048 | 0.720 ± 0.053 |

| | | |
|---|---|---|
| Final-20% hold-out (cluster CI), secondary | 0.810 (0.771–0.845) | 0.752 (0.694–0.806) |
| **Comparators** | | |
| SVM-RBF raw, hold-out (best classical hold-out, test-selected) | 0.842 (0.800–0.880) | 0.776 (0.712–0.832) |
| SVM-RBF raw, blocked CV / rolling origin | 0.755 ± 0.046 / 0.754 ± 0.085 | 0.685 ± 0.073 / 0.689 ± 0.090 |
| Amplitude fingerprint, LightGBM on \|x\|, rolling origin | 0.720 ± 0.074 | 0.648 ± 0.094 |
| Amplitude fingerprint, LR on \|x\|, rolling origin | 0.702 ± 0.104 | 0.627 ± 0.124 |
| Chance (uniform) | 0.250 | 0.333 |

Primary model fixed by the highest mean macro-F1 in nested purged blocked 5 × 2 CV on the development data (no test block used). Rolling origin: five folds, each testing a different 20% of every recording (400 rows per seizure class plus a fifth of the Bonn segments) with a 50-row purge; t-interval = mean ± $t_4 \cdot SD/\sqrt{5}$. Hold-out CI: class-stratified cluster bootstrap over 50-row chunks (2000 resamples). The SVM-RBF comparator is the best of about 16 classical configurations on the same final-20% block, i.e. selected on the test block. Per-fold values: Supplementary Tables S23–S24.

Errors under the blocked protocol were concentrated among the three seizure classes, especially seizure events (confusion matrices: Supplementary Fig. S14). The healthy class remained almost perfectly separable: the primary model's per-class F1 for class 0 was 0.999 (random) and 0.995 (blocked). In the binary healthy-versus-seizure task every non-baseline model reached macro-F1 ≥ 0.994 under both protocols (best 1.000 random and 0.999 blocked), and a single engineered feature with one threshold (within-row lag-1 autocorrelation) achieved a blocked-protocol accuracy of 0.997 (Supplementary Table S7).

### 4.5 Model comparison under repeated cross-validation and rolling origin

Table 4 compares the raw-value models in blocked CV and rolling origin, with Holm-adjusted p-values across all comparisons in the four task × protocol settings (all configurations, including engineered and raw+engineered features, random-protocol CV and hold-out scores: Supplementary Table S25). Under the blocked protocol, 11 of 21 four-class and 5 of 15 three-class comparisons with LightGBM (raw) remained significant. The top models were statistically indistinguishable: in the four-class task LightGBM (raw) could not be separated from 1D-CNN (raw), LightGBM (raw+eng), MLP (raw), RF (raw), RF (raw+eng), SVM-RBF (raw), Transformer (raw), XGBoost (raw), XGBoost (raw+eng), kNN (raw); in the three-class task from 1D-CNN (raw), BiLSTM (raw), LightGBM (raw+eng), MLP (raw), RF (raw),

SVM-RBF (raw), SVM-RBF (raw+eng), Transformer (raw), XGBoost (raw), XGBoost (raw+eng). Under the random protocol every comparison with kNN (raw) was significant. Fold-to-fold standard deviations of the classical models were much larger under the blocked protocol (four classes 0.030–0.057 versus 0.004–0.013). Tuning the deep models repaired the undertrained BiLSTM (four-class hold-out 0.468 with fixed defaults → 0.650 tuned), but no deep model was better than LightGBM (raw) in blocked CV (Supplementary Table S30). Friedman tests also rejected equal performance, but they are anti-conservative because folds share training data (descriptive Nemenyi ranks: Supplementary Fig. S5).

**Table 4. Macro-F1 of the raw-value models in blocked CV and rolling origin, with corrected tests**

| Model (raw values) | 4-class blocked CV | 4-class rolling origin | 4-class p (Holm) | 3-class blocked CV | 3-class rolling origin | 3-class p (Holm) |
|---|---|---|---|---|---|---|
| LightGBM | 0.777 ± 0.048 | 0.748 ± 0.080 | ref. | 0.720 ± 0.053 | 0.677 ± 0.104 | ref. |
| MLP† | 0.774 ± 0.054 | 0.739 ± 0.105 | 1.000 | 0.665 ± 0.074 | 0.693 ± 0.114 | 1.000 |
| XGBoost | 0.773 ± 0.048 | 0.739 ± 0.074 | 1.000 | 0.716 ± 0.049 | 0.674 ± 0.102 | 1.000 |
| RF | 0.763 ± 0.050 | 0.730 ± 0.089 | 1.000 | 0.693 ± 0.066 | 0.642 ± 0.115 | 1.000 |
| SVM-RBF | 0.755 ± 0.046 | 0.754 ± 0.085 | 0.938 | 0.685 ± 0.073 | 0.689 ± 0.090 | 1.000 |
| Transformer† | 0.749 ± 0.048 | 0.747 ± 0.088 | 1.000 | 0.705 ± 0.058 | 0.687 ± 0.122 | 1.000 |
| 1D-CNN† | 0.739 ± 0.047 | 0.678 ± 0.070 | 1.000 | 0.645 ± 0.099 | 0.594 ± 0.134 | 1.000 |
| kNN | 0.697 ± 0.030 | 0.685 ± 0.068 | 0.155 | 0.624 ± 0.072 | 0.600 ± 0.100 | 0.028 |
| BiLSTM† | 0.590 ± 0.038 | 0.640 ± 0.087 | <0.001 | 0.525 ± 0.111 | 0.432 ± 0.069 | 0.213 |
| LR | 0.352 ± 0.047 | 0.420 ± 0.060 | <0.001 | 0.389 ± 0.054 | 0.423 ± 0.047 | 0.002 |

Mean ± SD over 10 outer folds of purged blocked 5 × 2 CV or 5 rolling-origin folds. Rows sorted by four-class blocked CV. p: Nadeau–Bengio corrected resampled t-test against LightGBM (raw) (ref.), the best configuration in blocked CV, Holm-adjusted across all 72 comparisons in the four task × protocol settings. †Tuned by an inner purged blocked hold-out. All configurations, random-protocol CV and hold-out scores: Supplementary Table S25; all tests: Supplementary Table S33.

### 4.6 A recording amplitude fingerprint explains most of the leakage-aware performance

Table 5 compares the primary model with fingerprint and normalisation baselines (all baselines: Supplementary Fig. S15). A sign-free per-channel amplitude profile alone, |x| for the 16 channels, reached 0.720 ± 0.074 (four classes) and 0.648 ± 0.094 (three classes) in the rolling origin with LightGBM, and 0.702 ± 0.104 and 0.627 ± 0.124 with LR. Relative to chance, LightGBM(|x|) recovered 94.5% and 91.6% of the primary model's above-chance rolling-origin performance. In the pre-specified family of four tests, which compared the

primary model with the simplest fingerprint model, logistic regression on |x|, the primary model's advantage was 0.046–0.054 macro-F1 and not significant after Holm correction (p = 0.15–0.16). Removing each recording's per-channel offset and scale (per-recording z-scoring) reduced three-class performance to 0.572 (rolling origin) and 0.587 (blocked CV), and four-class performance by about 0.07. Per-row normalisation, which removes only the common offset and gain of each row, lost nothing (0.767 and 0.703), so what survives is mostly the relative spatial amplitude pattern of each recording, which, with one recording per class, cannot be separated from seizure type.

**Table 5. Primary model versus recording-fingerprint and normalisation baselines (macro-F1)**

| Input / model | 4-class rolling origin | 4-class blocked CV | 4-class hold-out | 3-class rolling origin | 3-class blocked CV | 3-class hold-out |
|---|---|---|---|---|---|---|
| Primary: LightGBM (raw) | 0.748 ± 0.080 | 0.777 ± 0.048 | 0.810 | 0.677 ± 0.104 | 0.720 ± 0.053 | 0.752 |
| Fingerprint: LightGBM on per-channel \|x\| | 0.720 ± 0.074 | 0.737 ± 0.049 | 0.778 | 0.648 ± 0.094 | 0.685 ± 0.054 | 0.726 |
| Fingerprint: LR on per-channel \|x\| | 0.702 ± 0.104 | 0.723 ± 0.059 | 0.775 | 0.627 ± 0.124 | 0.669 ± 0.050 | 0.727 |
| Row SD only (LR) | 0.568 ± 0.092 | 0.529 ± 0.089 | 0.647 | 0.490 ± 0.116 | 0.470 ± 0.105 | 0.588 |
| Row mean only (LightGBM) | 0.449 ± 0.030 | 0.434 ± 0.030 | 0.490 | 0.367 ± 0.043 | 0.372 ± 0.027 | 0.389 |
| Per-recording channel z-score (best: SVM-RBF) | 0.674 ± 0.032 | 0.704 ± 0.065 | 0.687 | 0.572 ± 0.042 | 0.587 ± 0.050 | 0.587 |
| Per-row normalisation (best: LightGBM) | 0.767 ± 0.044 | 0.784 ± 0.024 | 0.808 | 0.703 ± 0.049 | 0.705 ± 0.031 | 0.748 |
| Chance (uniform) | 0.250 | 0.250 | 0.250 | 0.333 | 0.333 | 0.333 |

Rolling origin: mean ± SD over 5 test blocks; blocked CV: mean ± SD over 10 folds of purged blocked 5 × 2 CV; hold-out: final-20% blocked test block. |x|: absolute value of each of the 16 columns. Normalised inputs were de-quantised with U(−0.5, 0.5) noise before normalisation (Section 3.6); per-recording z-scoring uses the class label to define the recording and is a diagnostic, not a deployable model. Pre-specified tests of the primary model against LR on |x|: Supplementary Table S27; all baselines with four models: Supplementary Table S26.

### 4.7 Temporal context: a separate, non-comparable setting

With temporal context, seizure types were more separable: on the same final-20% hold-out, multichannel band-power windows reached macro-F1 0.97 (95% cluster-bootstrap CI 0.90–1.00) versus 0.77 for a single-row SVM-RBF on the same rows (0.89 with per-window averaging), and in purged blocked CV 0.92 ± 0.10 versus 0.67 ± 0.08. Part of the gain is simple averaging (windowed minus window vote +0.086, paired CI -0.028 to 0.211).

Windows are a different unit of analysis, test sets are small and all windows of a class come from one recording, so this setting is not comparable with Tables 3–5 and is not evidence of patient-level generalisation (Supplementary Table S29, Fig. S17).

### 4.8 Calibration

Calibration deteriorated under the blocked protocol. On the four-class hold-out, the median ECE of the ten non-baseline models (over all feature sets) rose from 0.029 (random) to 0.093 (blocked). For the primary model, ECE rose from 0.010 to 0.077 (Brier score 0.025 → 0.276) in the four-class task and from 0.012 to 0.179 in the three-class task, i.e. the model became overconfident on temporally distant test blocks (reliability diagrams: Supplementary Fig. S4). Probabilities would therefore need recalibration on held-out recordings before any clinical use.

### 4.9 Interpretability and ablation

For the descriptive LightGBM (raw+engineered), the leading four-class features were t_hjorth_mob, X8, n_higuchi_fd, X11 under the random protocol and X11, X8, t_hjorth_mob, t_lag1_ac under the blocked protocol (Supplementary Fig. S8, Table S17). The healthy class was separated almost entirely by smoothness descriptors (Hjorth mobility, Higuchi fractal dimension, lag-1 autocorrelation), the source shortcut of Section 4.2, whereas the seizure classes were separated by raw values of particular columns (notably X11, X8 and X15), consistent with the amplitude fingerprint of Section 4.6. In the ablation (Supplementary Fig. S6), raw values alone outperformed all engineered features combined under the blocked protocol (0.763 vs 0.634 four-class CV macro-F1), because summary statistics computed across electrodes discard electrode identity.

### 4.10 Computational footprint on a general-purpose CPU

On one thread of the server-class Xeon CPU (Supplementary Table S31, Fig. S19), inference of the profiled models took 0.05–0.74 ms per sample in Python (0.02–0.09 ms with ONNX Runtime), and the serialised ONNX models occupied 16 kB to 2.3 MB. The profiled tuned LightGBM (73 features: 16 raw + 57 engineered) took 3.97 ms per sample, 87.5% of it Python feature extraction; the primary model (raw values only) needs no feature extraction. These figures were not measured on wearable or microcontroller hardware and do not establish edge feasibility; moreover, because healthy-class inputs are single-channel windows whereas seizure-class inputs are 16-channel instants, no single continuous input stream exists on which such a model could run.

## 5 Discussion

The main finding is that the public BEED file combines two data layouts and two data sources: the 'healthy' class consists of single-channel Bonn excerpts, whereas the seizure classes behave as multichannel time points of continuous recordings. This explains the near-perfect healthy-versus-seizure separation under both protocols, which reflects how the file was assembled, not evidence that seizures can be detected from a single row of EEG.

The second finding concerns evaluation. Consecutive seizure rows are 3.9 ms apart and highly correlated, so a random split evaluates interpolation between neighbouring samples; accuracies of 95–99% reported for row-level splits [8,10–12] agree with our random-protocol results. Under a pre-specified leakage-aware evaluation, four-class macro-F1 was about 0.75 rather than 0.98, the ranking of models changed, and the top models could not be separated statistically, in line with reports of leakage-inflated EEG results [18,19]. Model selection on a single test block adds a second, milder optimism: selecting the best of about 16 classical configurations on the favourable final-20% block would raise the estimate by about 0.09 (four classes) and 0.10 (three classes) relative to the pre-specified rolling-origin estimate.

Third, even the blocked protocol is optimistic about seizure-type discrimination, because training and test blocks come from the same recordings. With about 7.8 s per seizure class, recording identity and seizure type are confounded, and an amplitude fingerprint recovers about 90% of the above-chance performance. The informative spatial voltage pattern (Supplementary Fig. S8) is physiologically plausible, but with one recording per class it cannot be distinguished from recording-specific gain, offset and montage. Subject-independent evaluation is impossible with the public file, and claims of it should be treated with caution; future releases should provide segment-level multichannel data with subject identifiers, where the windowed analysis suggests temporal context helps.

**Answers to the research questions.** RQ1: the healthy class is identical to single-channel Bonn excerpts, and seizure-class rows behave as 16-channel time points rather than 200-time-point subject blocks. RQ2: random splits inflated macro-F1 from about 0.75 to 0.98 (four classes) and from 0.68 to 0.98 (three seizure types). RQ3: seizure classes were separated mainly by the per-recording spatial amplitude pattern, not by demonstrable seizure-type physiology, and probabilities were poorly calibrated on distant test blocks. RQ4: inference took 0.05–0.74 ms per sample on one server CPU thread; edge feasibility was not tested.

## 6 Conclusion

The public BEED file is not a homogeneous 16-channel dataset: its healthy class is identical to excerpts of the Bonn eyes-closed set, and its seizure classes are autocorrelated multichannel time points. Row-level random splits therefore inflate performance substantially. Under a pre-specified leakage-aware protocol, four-class macro-F1 was about 0.75 (rolling origin; 0.78 in blocked CV) rather than 0.98, and seizure-type macro-F1 about 0.68 (0.72) rather than 0.98, with most of the remaining signal attributable to recording-specific amplitude patterns. We recommend that BEED studies (i) exclude or replace the healthy class for clinical claims, (ii) use grouped, temporally blocked evaluation with purging, (iii) report cluster-bootstrap

confidence intervals and calibration, (iv) state the evaluation protocol explicitly, and (v) pre-specify the model and report rolling-origin or CV estimates alongside any single hold-out.

## 7 Limitations and Future Work

The public file contains no subject identifiers, so no protocol can assess generalisation to new patients; the blocked and rolling-origin protocols measure generalisation to unseen time within the same recordings, and grouping class 0 by Bonn segment rather than by volunteer slightly favours that class. Our structural inferences about classes 1–3 rest on statistical evidence rather than documentation, and one candidate splice point (class 1, row 1900) cannot be excluded. The CHB-MIT search (246 of 686 files, all 24 subjects; Supplementary Section S7.5) recovered a positive control but did not test resampled, filtered or re-montaged derivatives or other corpora (TUSZ, Siena), so the source of classes 1–3 remains unknown. The primary-model selection rule was fixed after an initial exploratory analysis that included hold-out results; a fully nested selection check inside the rolling origin gave almost the same estimates, which limits but does not remove this concern. De-quantisation removed the value-lattice artefact of per-recording normalisation, but other artefacts of this kind cannot be fully excluded. The 1D-CNN, BiLSTM and transformer treat the 16 electrodes as a sequence, which is not a temporal order; their inner tuning used a single blocked hold-out, and under the random protocol they used fixed defaults. Grids were deliberately compact, and footprint measurements were made on a server CPU, not on edge hardware. Future work will apply the protocol to segment-level BEED recordings if released, test cross-dataset transfer and evaluate calibrated, patient-independent seizure-type classifiers.

## Data Availability Statement

BEED is publicly available from the UCI Machine Learning Repository (doi:10.24432/C5K33B) [7]. The Bonn-derived Epileptic Seizure Recognition file used for

the provenance check is available from its creators (doi:10.24432/C5G308) [21], and CHB-MIT from PhysioNet [6]. All derived results, figures and code are provided in Supplementary File S1 and will be deposited in a public repository upon acceptance (link withheld for double-blind review). Software: BEED_code_S1 (version as submitted; archived version: [DOI to be added]); language Python 3.13; platform Linux (any operating system with Python 3.13 should work); dependencies listed in requirements.txt (scikit-learn, XGBoost, LightGBM, PyTorch and others; versions in Section 3.9).

## Ethics Statement

This study analysed only publicly available, de-identified EEG data released for research use; no new data were collected and the authors had no access to identifying information. Ethics approval and consent to participate were therefore not required for this secondary analysis; consent for the original recordings is described by the data creators. Consent for publication: not applicable.

## Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

[Placeholder for the authors to finalise.] AI agents were used to write the analysis code, run the experiments and draft and edit the manuscript text. All content was reviewed, verified and edited before submission, and responsibility for the publication rests with the named author(s). All numerical results were produced by the released code.

## Declaration of Conflict of Interest

The authors declare no conflict of interest. [Authors to confirm.]

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